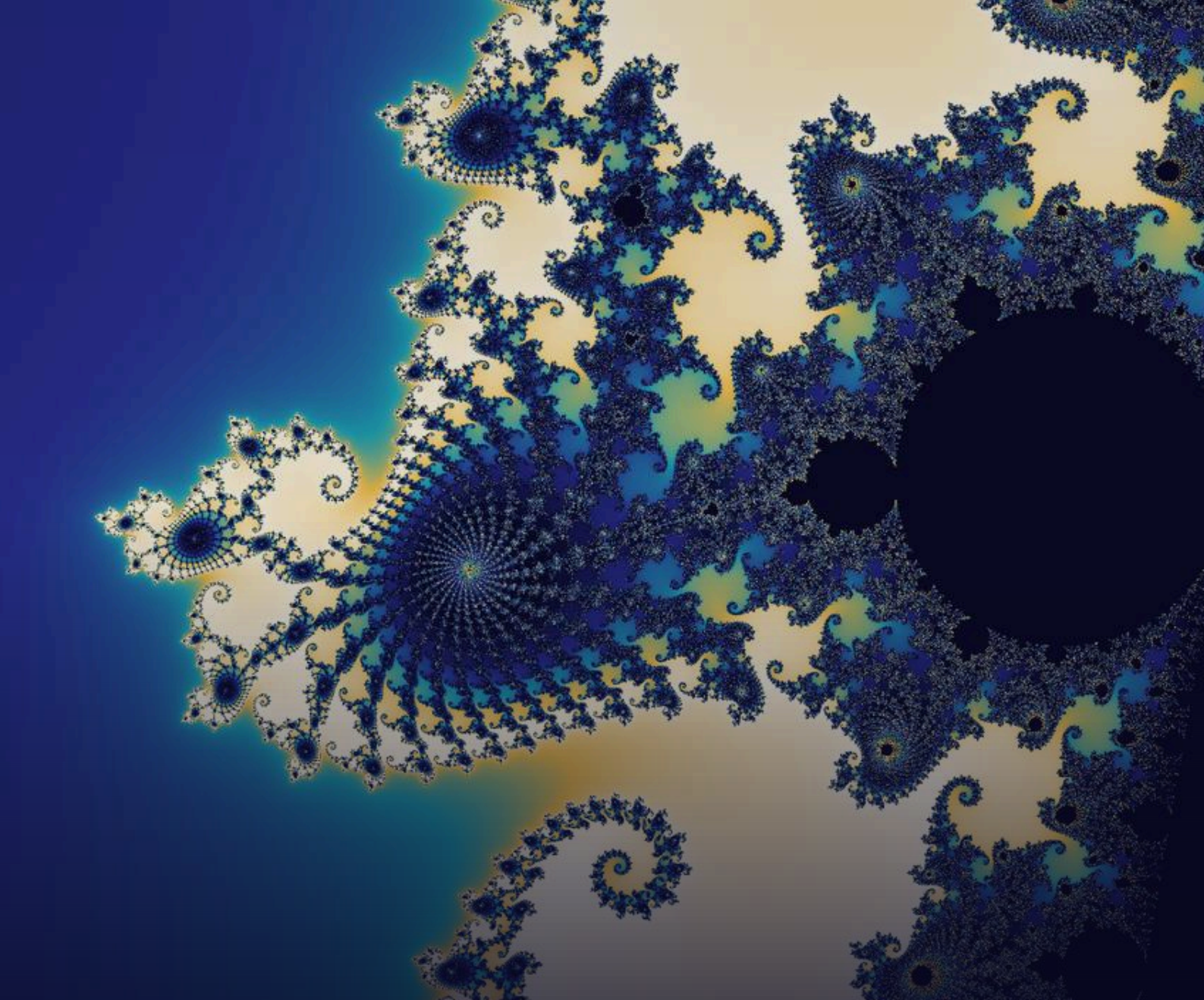


KINGSCLIFF HIGH SCHOOL · MATHEMATICS

MANDELBROT SERIES

Bridge to Year 9

Seven lessons to revise and strengthen the Year 8 number, algebra, Pythagoras and measurement skills that Year 9 is built on.

1 Number skills

2 Algebra basics

3 Expanding & factorising

4 Indices & square roots

5 Pythagoras' theorem

6 Equations & formulas

7 Area & volume

NAME _____ CLASS _____ TEACHER _____

How this booklet works

One lesson = three pages = three rounds

Each lesson starts with **key words**, **key ideas** and **examples** that your teacher works through with you, then three rounds of questions that get harder as you go.

- **Bronze – support.** Quick drill questions. Everyone starts here; most should be done without help.
- **Silver – core.** Multi-step questions and worded problems. Aim to finish this round.
- **Gold – challenge.** Harder problems that make you think. Try them once Silver is done.

Mixed questions

Questions with a chip like ↻ L2 come from an **earlier lesson**; chips like ↻ Percentages revisit **other Year 8 topics**. Mixing topics feels harder, but it makes what you learn last much longer.

Help along the way

Some Bronze questions have the **first step done for you** in grey – finish them off.

Calculators are allowed. Write down what you put into the calculator, and round only at the last step.

Write your working in the spaces provided.

Calculator tips

- Put negatives in brackets: $(-3)^2 = 9$ but $-3^2 = -9$.
- Use the fraction key for fractions and mixed numbers.
- Use the π key, not 3.14.
- Use x^2 and $\sqrt{\ }$ for Pythagoras: $\sqrt{(6^2 + 8^2)}$.
- Round only at the end, and say how you rounded (e.g. 1 d.p.).
- Estimate first, so you can spot a wrong key press.

My progress

LESSON	SYLLABUS	B	S	G	CONFIDENCE (CIRCLE)
1 Number skills	MA4-INT-C-01 · MA4-FRC-C-01	☐	☐	☐	1 2 3 4
2 Algebra basics	MA4-ALG-C-01	☐	☐	☐	1 2 3 4
3 Expanding & factorising	MA4-ALG-C-01	☐	☐	☐	1 2 3 4
4 Indices & square roots	MA4-IND-C-01	☐	☐	☐	1 2 3 4
5 Pythagoras' theorem	MA4-PYT-C-01	☐	☐	☐	1 2 3 4
6 Equations & formulas	MA4-EQU-C-01	☐	☐	☐	1 2 3 4
7 Area & volume	MA4-ARE-C-01 · MA4-VOL-C-01	☐	☐	☐	1 2 3 4

Pre-test / 20 | Post-test / 20

A 50-minute lesson

0–5	5–18	18–40	40–50
Key ideas & examples	Bronze round	Silver round	Gold · check answers

Finished early? Go straight to Gold. Stuck? Look back at the examples you did with your teacher, then ask.

Getting the most out of it

- Try every question before asking for help – a wrong attempt teaches more than a blank.
- Show your working: in Year 9 many marks are for method.
- Check answers with your teacher; circle any you got wrong and redo them a few days later.
- Mixed (↻) questions are meant to make you stop and think "which skill is this?"

Why Mandelbrot?

Every banner in this booklet is a zoom into the Mandelbrot set, drawn by repeating one tiny rule, $z \to z^2 + c$, over and over. Simple algebra, repeated carefully, builds something extraordinary – which is exactly what Year 9 will ask of you.

BRIDGE TO YEAR 9

Pre-test

Do this before Lesson 1. It shows what you already know and which rounds to focus on.

20 marks · 20 minutes · Calculator allowed

Name _____ Date _____ **Score / 20**

1 Number skills

Lesson 1 · 3 marks

a) Evaluate $-8 + 3 \times (-2)$ [1]

b) Evaluate $\frac{2}{3} + \frac{1}{4}$ [1]

c) Increase $\$80$ by 15% [1]

2 Algebra basics

Lesson 2 · 3 marks

a) Simplify $5x + 3y - 2x$ [1]

b) Simplify $4a \times 3b$ [1]

c) Evaluate $2p + q$ when $p = -3$ and $q = 4$ [1]

3 Expanding & factorising

Lesson 3 · 3 marks

a) Expand $3(x + 4)$ [1]

b) Expand and simplify $2(a + 3) + 4a$ [1]

c) Factorise $12m + 18$ [1]

4 Indices

Lesson 4 · 3 marks

a) Write $5^7 \times 5^2 \div 5^6$ as a single power [1]

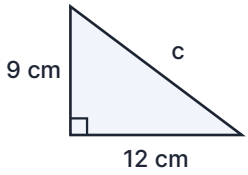
b) Evaluate $2^3 \times 3^2$ [1]

c) Write 180 as a product of prime factors [1]

5 Pythagoras' theorem

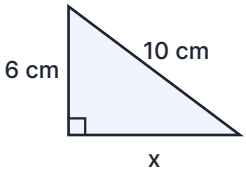
Lesson 5 · 3 marks

a) Find c.



[1]

b) Find x.



[1]

c) Find the hypotenuse if the shorter sides are 4 cm and 7 cm (1 d.p.)

[1]

6 Solve

Lesson 6 · 3 marks

a) Solve $4x - 7 = 13$

[1]

b) Solve $5a + 3 = 2a + 9$

[1]

c) Solve $2x^2 = 72$

[1]

7 Area and volume

Lesson 7 · 2 marks

a) Find the area of a trapezium with parallel sides 5 cm and 9 cm and height 4 cm.

[1]

b) Find the volume of a cylinder with radius 3 cm and height 5 cm (1 d.p.).

[1]

Where should I start each lesson?

QUESTION	LESSON	MY MARKS	START AT…
1 Number skills	Lesson 1	/ 3	Bronze ☐ Silver ☐ Gold ☐
2 Algebra basics	Lesson 2	/ 3	Bronze ☐ Silver ☐ Gold ☐
3 Expanding & factorising	Lesson 3	/ 3	Bronze ☐ Silver ☐ Gold ☐
4 Indices	Lesson 4	/ 3	Bronze ☐ Silver ☐ Gold ☐
5 Pythagoras' theorem	Lesson 5	/ 3	Bronze ☐ Silver ☐ Gold ☐
6 Solve	Lesson 6	/ 3	Bronze ☐ Silver ☐ Gold ☐
7 Area and volume	Lesson 7	/ 2	Bronze ☐ Silver ☐ Gold ☐

Full marks on a question? Move quickly through Bronze and spend your time on Silver and Gold. Lost marks? Do every Bronze question first.

LESSON 1 OF 7

Number skills

Integers · order of operations · fractions · decimals · percentages · rounding

50 min

Calculator allowed

Outcomes: MA4-INT-C-01 · MA4-FRC-C-01 · MAO-WM-01 | Gold: harder Stage 4 problems (no Stage 5 content) *NCM 8 Ch 2, Ch 6*

KEY WORDS

integer a whole number: positive, negative or zero

numerator the top number of a fraction

denominator the bottom number of a fraction

percentage a number out of 100

round give an approximate value, e.g. to 1 decimal place

KEY IDEAS

- **Integers:** $\times$ or $\div$ with the same signs gives a positive; different signs gives a negative. Subtracting a negative is the same as adding: $4 - (-2) = 6$.
- **Order of operations:** brackets, then $\times$ and $\div$ (left to right), then $+$ and $-$ (left to right).
- **Fractions:** to add or subtract, use a common denominator. To multiply, multiply tops and bottoms. To divide, flip the second fraction and multiply.
- **Percentages:** 15% of $60 = 0.15 \times 60 = 9$. Increase by 15%: $\times 1.15$. Decrease by 20%: $\times 0.8$.
- **Rounding:** look at the next digit. 5 or more rounds up: $3.476 \approx 3.5$.

EXAMPLES – your teacher will work these with you

Example 1 Evaluate $-6 + 4 \times (-3)$.

Example 2 Evaluate $\frac{2}{3} + \frac{3}{4}$.

Example 3 Increase $\$240$ by 15%.

B Bronze round

Support · everyone starts here

Evaluate.

1 $-5 + 8$

2 $-3 - 6$

3 $4 - (-2)$

4 -4×5

5 $-18 \div (-3)$

6 $3 + 4 \times 2$

7 $(3 + 4) \times 2$

8 $\frac{1}{4} + \frac{2}{4}$

9 $\frac{1}{3} + \frac{1}{4}$

$= \frac{4}{12} + \frac{3}{12} = ___$

10 $\frac{2}{3} \times \frac{3}{5}$

11 15% of 60

$= 0.15 \times 60 = ___$

12 25% of 80

13 Round 3.476 to 1 decimal place

14 Write 0.6 as a percentage

15 Simplify the ratio $6 : 9$

↻ Ratios

16 Two angles on a straight line: one is 130°. Find the other.

↻ Angles

S Silver round

Core · aim to finish this round

17 Evaluate $-2 \times (-3) \times (-4)$

18 Evaluate $20 - 3 \times (-4)$

19 Evaluate $\frac{3}{4} - \frac{1}{3}$

20 Evaluate $1\frac{1}{2} + 2\frac{2}{3}$

21 Evaluate $\frac{2}{5} \div \frac{4}{15}$

22 Evaluate $2\frac{1}{4} \times 1\frac{1}{3}$

23 Increase $\$80$ by 15%

24 Decrease $\$120$ by 30%

25 Write 18 as a percentage of 72

26 Evaluate 2.4×0.3

27 Evaluate $7.2 \div 0.4$

28 Round 12.6849 to 2 decimal places

29 Share $\$60$ in the ratio $1 : 2$

↻ Ratios

30 Find the mean of 3, 5 and 10

↻ Data

31 A $\$95$ pair of thongs is reduced by 20%. Find the sale price.

32 At 6 am the temperature at Thredbo is -4°C. By noon it has risen 11°, then it falls 3° by evening. What is the evening temperature?

G Gold round

Challenge · stretch yourself

33 A shop increases a price by 10%, then reduces the new price by 10%. Is the final price the same as the original? Use $\$100$ to explain.

34 What fraction is exactly halfway between $\frac{1}{3}$ and $\frac{1}{2}$?

35 After a 15% discount a jacket costs $\$68$. What was the original price?

36 Evaluate $-3 \times (4-7)^2 \div 9$.

37 Two integers add to -2 and multiply to -15. Find them.

38 Which is the better buy: 500 g of coffee for $\$12.50$ or 750 g for $\$17.40$? Show working.

LESSON 2 OF 7

Algebra basics

Writing expressions · like terms · multiplying and dividing terms · substitution

50 min

Calculator allowed

Outcomes: MA4-ALG-C-01 · MAO-WM-01 | Gold: harder Stage 4 problems (no Stage 5 content) *NCM 8 Ch 3.01–3.05*

KEY WORDS

pronumeral a letter that stands for a number

term part of an expression, e.g. $5x$ or $-3y$

like terms same pronumerals with the same powers

coefficient the number in front of a pronumeral

substitute replace a pronumeral with a number

KEY IDEAS

- **Like terms** have exactly the same pronumerals: $3xy$ and $-5xy$ are like terms; $3x$ and $3x^2$ are not.
- **Add or subtract** only like terms: $5a + 2b + 3a = 8a + 2b$.
- **Multiply** the numbers, then the pronumerals: $4a \times 3b = 12ab$, $3m \times m = 3m^2$.
- **Divide** by writing a fraction and cancelling: $12xy \div 4x = \dfrac{12xy}{4x} = 3y$.
- **Substitute** using brackets: if $a = -3$, then $2a = 2 \times (-3) = -6$ and $a^2 = (-3)^2 = 9$.

EXAMPLES – your teacher will work these with you

Example 1 Simplify $5x + 3y - 2x + 7y$.

Example 2 Simplify $4a \times 3ab$.

Example 3 Evaluate $2a^2 - 3b$ when $a = -3$ and $b = 4$.

B Bronze round

Support · everyone starts here

Simplify, or answer as instructed.

1 $7a + 3a$

2 $9m - 4m$

3 $5x + 2y + 3x$

4 $4 \times 3k$

5 $5a \times 2b$

6 $12y \div 3$

7 $20ab \div 5b$

8 $6p - p + 2q$

9 $3m \times m$

10 $3a + 4b + 2a + b$
$= (3a + 2a) + (4b + b) =$ ___

11 If $x = 4$: $3x + 5$
$= 3 \times 4 + 5 =$ ___

12 Write an expression for "5 more than n"

13 Write an expression for "the product of 3 and y"

14 Evaluate $-7 + 12$

↻ L1

15 $\frac{3}{4}$ of 36

↻ L1

16 If $a = 3$, $b = -4$: $2a + b$

S Silver round

Core · aim to finish this round

Simplify.

17 $3x^2 + 5x - x^2 + 2x$

18 $4ab - 3ba + 2a$

19 $2xy \times 3x$

20 $15a^2b \div 5ab$

21 $\dfrac{8xy}{12y}$

22 $-6m \times (-2n)$

If $p = -3$ and $q = 5$, evaluate:

23 $p^2 - 2q$

24 $(p + q)^2$

25 $\dfrac{q - p}{4}$

26 $2pq$

27 Increase $\$60$ by 15%

L1

28 If $y = 3x - 5$, find y when $x = -2$

Linear

29 Pizzas cost $\$14$ each, plus a $\$6$ delivery fee.
a) Write an expression for the cost of n pizzas.
b) Find the cost of 3 pizzas.

30 Mia says $3x + 2y = 5xy$. Choose values for x and y to show she is wrong.

G Gold round

Challenge · stretch yourself

31 The nth term of a pattern is $3n + 2$.
a) Find the 50th term. b) Is 100 a term? Explain.

32 A rectangle is $2a$ cm long and a cm wide. Write and simplify expressions for its perimeter and its area.

33 Evaluate $a^2 - 2ab + b^2$ when $a = 7.5$ and $b = 2.5$.

34 A taxi charges $\$4$ plus $\$2.50$ per km. Write an expression for the fare for k km, then find the fare for 12 km.

35 Find a value of x that makes x^2 smaller than x.

36 Simplify $\dfrac{6x}{5} \times \dfrac{10}{3x}$.

LESSON 3 OF 7

Expanding & factorising

The distributive law · expanding and simplifying · highest common factor · factorising

50 min

Calculator allowed

Outcomes: MA4-ALG-C-01 · MAO-WM-01 | Gold: harder Stage 4 problems (no Stage 5 content) *NCM 8 Ch 3.07–3.09*

KEY WORDS

- **expand** remove the brackets by multiplying
- **distributive law** $a(b+c) = ab + ac$
- **factor** a number or term that divides exactly into another
- **HCF** highest common factor
- **factorise** write as a product using brackets, e.g. $3(x+4)$

KEY IDEAS

- **Expand:** multiply *every* term inside the brackets by the term outside: $3(x+5) = 3x + 15$.
- **Expand, then collect like terms:** $2(x+3) + 4x = 2x + 6 + 4x = 6x + 6$.
- **Factorise** is the reverse of expand: $6x + 15 = 3(2x+5)$.
- Find the **HCF** of the numbers and the pronumerals, write it outside the bracket, then divide each term by it.
- **Check** your factorising by expanding your answer.

EXAMPLES – your teacher will work these with you

Example 1 Expand $4(2a+3)$.

Example 2 Expand and simplify $3(x+2) + 2(x+5)$.

Example 3 Factorise $12x + 18$.

B Bronze round — Support · everyone starts here

Expand.

1 $3(x+4)$

2 $5(a-2)$

3 $2(3m+1)$

4 $4(x+5)$

$= 4 \times x + 4 \times 5 = ___$

5 $x(x+6)$

6 $a(b-3)$

7 Evaluate $-15 \div (-3)$ L1

8 Simplify $3a \times 4a$ L2

Find the HCF, then factorise.

9 HCF of 12 and 18

10 $3x + 12$

11 $5a - 20$

12 $7y - 14$

13 $6x + 9$

$\text{HCF} = 3: \ 3(__ + __)$

14 $x^2 + 3x$

S Silver round

Core · aim to finish this round

Expand and simplify.

15 $2a(3a + 5b)$

16 $5m(2 - m)$

17 $3(x + 2) + 4(x + 1)$

18 $x(x + 3) + 2(x - 4)$

19 $2(3y + 4) + 5y$

20 Evaluate $\frac{2}{3} + \frac{1}{6}$

L1

Factorise fully.

21 $4m + 6$

22 $x^2 - 7x$

23 $4a^2 + 10a$

24 $9mn - 6m$

25 $15p^2q + 20pq^2$

26 $3x^2 + 6x + 9$

27 20% of $\$85$

L1

28 If $x = -2$: $3x^2 - x$

L2

29 Jack wrote $2(x + 5) = 2x + 5$. Find his mistake and correct it.

30 A rectangle has area $(6x + 15)$ cm^2 and width 3 cm. Find an expression for its length.

31 Use factorising to work out $17 \times 23 + 17 \times 77$ in your head.

G Gold round

Challenge · stretch yourself

32 Expand and simplify $2(3x + 1) + 3(x + 4) - 5$, then factorise your answer.

33 Factorise fully $12a^2b + 18ab^2 - 6ab$.

34 Write three different expressions, each with brackets, that expand to $12x + 8$.

35 Factorise $n^2 + n$. Use your answer to explain why $n^2 + n$ is always even for whole numbers n.

36 A garden is $(x + 4)$ m long and 3 m wide. A path 1 m wide runs along one long side. Write and expand an expression for the area of the garden *without* the path.

37 Sam factorised $8x^2 - 12x$ as $2x(4x - 6)$. Is this fully factorised? Explain and correct it.

LESSON 4 OF 7

Indices & square roots

Index notation · index laws · the zero index · prime factors · square and cube roots

50 min

Calculator allowed

Outcomes: MA4-IND-C-01 · MAO-WM-01 | Gold: harder Stage 4 problems (no Stage 5 content) *NCM 8 Ch 2.08, 2.10–2.12*

KEY WORDS

base the number being multiplied: 2 in 2^5

index how many times it is multiplied: 5 in 2^5

prime a number with exactly two factors, e.g. $2, 3, 5, 7$

square root $\sqrt{49} = 7$ because $7^2 = 49$

cube root $\sqrt[3]{8} = 2$ because $2^3 = 8$

KEY IDEAS

- $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$. The **base** is 2 and the **index** is 5.
- **Multiply:** add indices, $a^m \times a^n = a^{m+n}$. **Divide:** subtract, $a^m \div a^n = a^{m-n}$.
- **Power of a power:** multiply, $(a^m)^n = a^{mn}$. **Zero index:** $a^0 = 1$.
- These laws only work when the **bases are the same**.
- **Prime factors:** use a factor tree, then index form: $72 = 2^3 \times 3^2$.

EXAMPLES – your teacher will work these with you

Example 1 Evaluate 3^4.

Example 2 Write $2^3 \times 2^4$ as a single power.

Example 3 Write 72 as a product of prime factors.

B Bronze round

Support · everyone starts here

Evaluate, or answer as instructed.

1 Write $5 \times 5 \times 5$ in index form

2 2^5

3 10^3

4 $3^2 + 4^2$

5 7^0

6 $\sqrt{81}$

7 $\sqrt[3]{64}$

8 Single power: $2^3 \times 2^4$

$= 2^{3+4} =$ ___

9 Single power: $5^9 \div 5^3$

10 Single power: $(3^2)^4$

11 Simplify $4x \times 5y$ L2

12 Factorise $8x - 20$ L3

13 Write 36 as a product of primes in index form

$36 = 2 \times 2 \times 3 \times 3 =$ ___

14 Write 24 as a product of primes in index form

15 $\frac{1}{2} + \frac{1}{5}$ L1

S Silver round

Core · aim to finish this round

Evaluate (use the index laws first).

16 $7^4 \times 7^6 \div 7^8$

17 $(2^4)^3 \div 2^{10}$

18 $\dfrac{3^5 \times 3^4}{3^7}$

19 $4^0 + 4^1 + 4^2$

20 $5 \times 2^3 - (3+1)^2$

21 $2^3 \times 3^2$

22 Find n: $2^n = 128$

23 Find n: $3^n \times 3^2 = 3^9$

24 90 as a product of primes

25 540 as a product of primes

26 $\sqrt{50}$ lies between which two whole numbers?

27 $\sqrt{50}$ to 1 decimal place

28 Decrease $\$250$ by 20%

L1

29 Expand $3(2x+5)$

L3

30 True or false: $2^3 \times 2^4 = 4^7$? Explain.

31 One bacterium doubles every hour. How many are there after 10 hours? Write your answer as a power and as a number.

32 Explain why $5^0 = 1$, using $5^3 \div 5^3$.

G Gold round

Challenge · stretch yourself

33 Which is larger, 2^{30} or 3^{20}? Hint: $2^{30} = (2^3)^{10}$.

34 Find n if $4^n = 2^{10}$. (Write 4 as a power of 2.)

35 Use $1764 = 2^2 \times 3^2 \times 7^2$ to find $\sqrt{1764}$ without a calculator.

36 What is the smallest whole number you can multiply 72 by to make a perfect square?

37 Evaluate $\dfrac{2^{10}}{2^7 \times 2^3} + 6^0$.

38 A square has area $200\ \text{cm}^2$. Find its side length correct to 1 decimal place.

LESSON 5 OF 7

Pythagoras' theorem

The hypotenuse · finding the hypotenuse · finding a shorter side · testing for a right angle · triads

50 min

Calculator allowed

Outcomes: MA4-PYT-C-01 · MAO-WM-01 | Gold: harder Stage 4 problems (no Stage 5 content) *NCM 8 Ch 1*

KEY WORDS

hypotenuse the longest side, opposite the right angle

right angle an angle of 90°, marked with a small square

exact answer leaves the square root, e.g. $\sqrt{74}$

approximate a rounded decimal, e.g. 8.60

triad three whole numbers with $a^2 + b^2 = c^2$

KEY IDEAS

- The **hypotenuse** c is the longest side, opposite the right angle. **Pythagoras:** $c^2 = a^2 + b^2$.
- **Finding the hypotenuse:** square, **add**, then take the square root.
- **Finding a shorter side:** square, **subtract**, then take the square root: $a^2 = c^2 - b^2$.
- Round only at the **last step**, e.g. $c = \sqrt{74} \approx 8.60$.
- **Testing:** if $a^2 + b^2 = c^2$ (with c the longest side), the triangle is right-angled.

EXAMPLES – your teacher will work these with you

Example 1 Find c if the shorter sides are 5 cm and 12 cm.

Example 2 Find x if the hypotenuse is 9 cm and one side is 5 cm (1 d.p.).

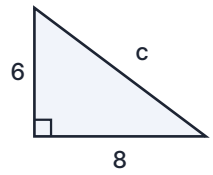
Example 3 Is a triangle with sides $7, 24, 25$ right-angled?

B Bronze round

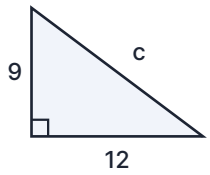
Support · everyone starts here

Find the hypotenuse c. Round to 1 decimal place where needed. (Lengths in cm.)

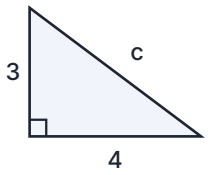
1



2

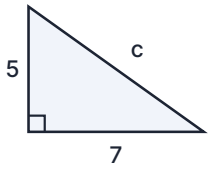


3



$c^2 = 9 + 16 = ___$

4



5 In $\triangle PQR$, $\angle Q = 90^\circ$. Name the hypotenuse.

6 Evaluate $\sqrt{49}$

L4

7 Round 8.6023 to 2 decimal places

L1

8 Evaluate $6^2 + 8^2$

9 Hypotenuse: shorter sides 5 cm and 12 cm

10 Hypotenuse: shorter sides 2 cm and 3 cm (1 d.p.)

11 Find x: hypotenuse 10 cm, one side 6 cm

$x^2 = 100 - 36 = ___$

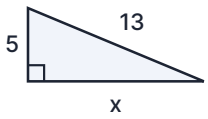
12 Evaluate $\sqrt{7^2 + 24^2}$

S Silver round

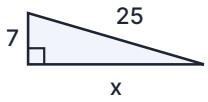
Core · aim to finish this round

Find x. Round to 1 decimal place where needed. (Lengths in cm.)

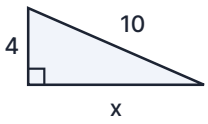
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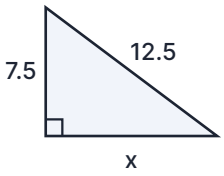
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15



16



17 Is a triangle with sides $9, 40, 41$ right-angled?

18 Is a triangle with sides $6, 9, 11$ right-angled?

19 Is $20, 21, 29$ a Pythagorean triad?

20 Factorise $x^2 + 4x$

L3

21 Evaluate $5^0 + 2^4$

L4

22 If $a = -3$, $b = 4$: $a^2 + b^2$

L2

23 Shorter side: hypotenuse 15 cm, one side 9 cm

24 Diagonal of a rectangle 30 cm by 40 cm

25 20% discount on $\$45$: sale price?

L1

26 A 5 m ladder leans against a wall with its foot 1.4 m from the wall. How high up the wall does it reach?

27 A TV screen is 80 cm wide and 45 cm high. Find the length of its diagonal to 1 decimal place.

28 Ava walks 600 m east along Marine Parade, Kingscliff, then 250 m north. How far is she in a straight line from where she started?

G Gold round

Challenge · stretch yourself

29 A 3 m ladder must reach 2.8 m up a wall. How far from the wall should its foot be? (1 d.p.)

30 Find the perimeter of a right-angled triangle whose shorter sides are 6 cm and 8 cm.

31 An isosceles triangle has equal sides of 13 cm and a base of 10 cm. Find its height and its area.

32 A rectangular paddock is 120 m by 50 m. How much shorter is it to walk across the diagonal than around two sides?

33 $3, 4, 5$ is a triad. Write two more triads that are multiples of it and show they work.

34 A kite string is 50 m long. The kite is directly above a point 30 m from where Leo stands. How high is the kite? (Ignore Leo's height.)

LESSON 6 OF 7

Equations & formulas

Inverse operations · one- and two-step equations · unknowns on both sides · $x^2 = c$ · formulas

50 min

Calculator allowed

Outcomes: MA4-EQU-C-01 · MAO-WM-01 | Gold: harder Stage 4 problems (no Stage 5 content) *NCM 8 Ch 9*

KEY WORDS

equation a number sentence with an equals sign, e.g. $2x + 3 = 11$

solution the value that makes the equation true

inverse operation the opposite operation, e.g. $+$ and $-$

formula a rule linking quantities, e.g. $A = lb$

check substitute your answer back into the equation

KEY IDEAS

- Keep the equation **balanced**: do the same inverse operation to both sides.
- **Two steps:** undo $+$ or $-$ first, then $\times$ or $\div$: $2x + 3 = 11 \Rightarrow 2x = 8 \Rightarrow x = 4$.
- **Unknowns on both sides:** first move the pronumeral terms to one side.
- $x^2 = c$ has **two** solutions: $x = \pm\sqrt{c}$, because $(-6)^2 = 6^2 = 36$.
- **Formulas:** substitute the values you know, then solve. Always **check** by substituting back.

EXAMPLES – your teacher will work these with you

Example 1 Solve $5x - 7 = 18$.

Example 2 Solve $7x - 3 = 4x + 9$.

Example 3 Solve $2x^2 = 50$.

B Bronze round

Support · everyone starts here

Solve.

1 $x + 9 = 15$

2 $y - 4 = -3$

3 $6m = 42$

4 $\frac{k}{5} = 3$

5 $2x + 3 = 11$

6 $3a - 5 = 16$

7 $4x + 7 = 31$

$4x =$ ___, $x =$ ___

8 $\frac{n}{4} + 2 = 5$

9 $x^2 = 36$

10 $10 - x = 4$

11 Expand $6(x - 2)$ ↻ L3

12 Evaluate $3^2 \times 3^3$ ↻ L4

13 Hypotenuse if the shorter sides are 3 and 4 ↻ L5

14 $5x = 2x + 12$

$3x = 12$, $x =$ ___

S Silver round

Core · aim to finish this round

Solve. Round to 2 decimal places where needed.

15 $6x - 5 = 2x + 11$

16 $5a + 2 = 2a - 10$

17 $\dfrac{3x}{4} = 9$

18 $8 - 3x = 2x - 7$

19 $0.5x + 4 = 7$

20 $\dfrac{x - 4}{3} = 5$

21 $3x^2 = 48$

22 $5x^2 = 40$

23 30% of $\$45$

↻ L1

24 Evaluate $\dfrac{5}{6} - \dfrac{1}{4}$

↻ L1

Substitute into the formula, then solve.

25 $v = u + at$. Find t when $v = 25$, $u = 5$, $a = 4$.

26 $A = \frac{1}{2}bh$. Find h when $A = 30$ and $b = 8$.

27 $A = \pi r^2$. Find r ($r > 0$) when $A = 50$. (2 d.p.)

28 $F = 1.8C + 32$. Find C when $F = 95$.

29 I think of a number, multiply it by 4 and subtract 7. The answer is 2 times the number plus 9. Find the number.

30 A Tweed Heads taxi charges $\$4.50$ plus $\$2.20$ per km. A trip cost $\$37.50$. How long was the trip?

G Gold round

Challenge · stretch yourself

31 A plumber charges $\$80$ to come out plus $\$65$ per hour. A bill is $\$307.50$. How many hours did the plumber work?

32 The length of a rectangle is 3 cm more than its width. Its perimeter is 42 cm. Find its length and width.

33 Solve $x^2 + 5 = 86$.

34 Three consecutive whole numbers add to 138. Find them.

35 A circle has area $200\ \text{cm}^2$. Find its radius to 1 decimal place.

36 Two phone plans: Plan A costs $\$20$ plus $\$0.50$ per GB; Plan B costs $\$12$ plus $\$1$ per GB. For how many GB do they cost the same?

LESSON 7 OF 7

Area & volume

Rectangles, triangles, parallelograms, trapeziums, kites · circles · composite areas · prisms and cylinders

50 min

Calculator allowed

Outcomes: MA4-ARE-C-01 · MA4-VOL-C-01 · MAO-WM-01 | Gold: harder Stage 4 problems (no Stage 5 content) *NCM 8 Ch 5*

KEY WORDS

perpendicular height the height at right angles to the base

composite shape a shape made of simpler shapes

cross-section the face that is the same all the way through a prism

prism a solid with a uniform cross-section

capacity how much a container holds: $1000\ \text{cm}^3 = 1\ \text{L}$

KEY IDEAS

- $A_{\text{rectangle}} = lb$ $A_{\text{triangle}} = \frac{1}{2}bh$ $A_{\text{parallelogram}} = bh$ (h is the **perpendicular** height).
- $A_{\text{trapezium}} = \frac{h}{2}(a+b)$ $A_{\text{kite}} = \frac{1}{2}xy$ $A_{\text{circle}} = \pi r^2$.
- **Composite shapes:** split into simple shapes and add, or take a piece away and subtract.
- **Prism:** $V = Ah$ (A = area of the cross-section). **Cylinder:** $V = \pi r^2 h$.
- Units: cm^2 for area, cm^3 for volume; $1\ \text{cm}^3 = 1$ mL and $1\ \text{m}^3 = 1000$ L.

EXAMPLES – your teacher will work these with you

Example 1 Trapezium: $a = 6$, $b = 10$, $h = 5$ (cm). Find the area.

Example 2 Circle: radius 3 cm. Find the area (1 d.p.).

Example 3 Cylinder: $r = 3$ cm, $h = 10$ cm. Find the volume (1 d.p.).

B Bronze round

Support · everyone starts here

Find the area or volume. Include units. Round to 1 decimal place where needed.

1 Rectangle 9 m by 4 m

2 Triangle: base 10 cm, height 6 cm

3 Parallelogram: base 12 cm, height 5 cm

4 Trapezium: $a = 4$, $b = 8$, $h = 5$ (cm)

5 Kite: diagonals 6 cm and 10 cm

6 Circle: radius 5 cm

7 Prism: cross-section $15\ \text{cm}^2$, height 8 cm

$V = 15 \times 8 =$ ___

8 Rectangular prism $5 \times 4 \times 3$ (cm)

9 Cube of side 6 cm

10 Solve $3x + 4 = 19$

↻ L6

11 Hypotenuse: shorter sides 6 and 8

↻ L5

12 Evaluate 4^3

↻ L4

13 Square of side 7 cm

14 Circle: diameter 8 cm

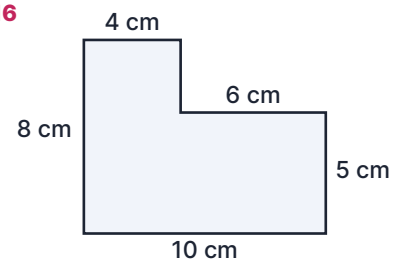
15 Cylinder: $r = 1$ cm, $h = 10$ cm

S Silver round

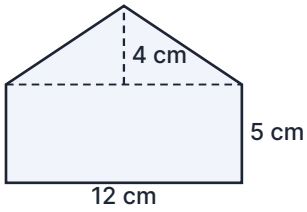
Core · aim to finish this round

Find the area of the first two shapes (drawn to scale) and the volume of the prism.

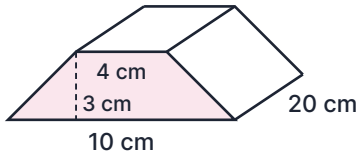
16



17



18



Volume? (not to scale)

Round to 1 decimal place where needed.

19 Semicircle: diameter 10 cm. Area?

20 Cylinder: $r = 5$ cm, $h = 8$ cm. Volume?

21 Cylinder: diameter 10 cm, $h = 12$ cm. Volume?

22 Parallelogram: area $72\ \text{cm}^2$, base 9 cm. Height?

↻ L6

23 A pool is 25 m by 10 m by 1.5 m deep. Volume, and capacity in litres?

24 Rectangle 4 by $(x + 3)$. Area, expanded?

↻ L3

25 Rhombus: diagonals 12 m and 7 m. Area?

26 Triangular prism: triangle base 6 cm, height 4 cm; length 10 cm. Volume?

27 Cube with volume $343\ \text{cm}^3$. Side length?

↻ L4

28 A garden bed is a trapezium with parallel sides 3.5 m and 5.5 m, 2 m apart. It is filled with soil 0.3 m deep. Find the volume of soil.

29 Explain why the area of a triangle is half the area of a rectangle with the same base and height.

G Gold round

Challenge · stretch yourself

30 A pipe is 100 cm long, with outer radius 5 cm and inner radius 4 cm. Find the volume of material (1 d.p.).

31 A cylinder has volume $1000\ \text{cm}^3$ and radius 5 cm. Find its height (1 d.p.).

32 A 10 cm cube has a cylindrical hole of radius 2 cm drilled straight through it. Find the volume left (1 d.p.).

33 A cylindrical rainwater tank has radius 1.2 m and height 2 m. How many litres does it hold? (nearest litre)

34 A trapezium has parallel sides x cm and $(x+4)$ cm, height 5 cm and area $45\ \text{cm}^2$. Find x.

35 What happens to the area of a circle when its radius is doubled? Explain.

BRIDGE TO YEAR 9

Post-test

Do this after Lesson 7. Compare your score with your pre-test.

20 marks · 20 minutes · Calculator allowed

Name _____ Date _____ **Score / 20**

1 Number skills

Lesson 1 · 3 marks

a) Evaluate $-10 + 2 \times (-4)$

[1]

b) Evaluate $\frac{3}{5} + \frac{1}{2}$

[1]

c) Decrease $\$90$ by 20%

[1]

2 Algebra basics

Lesson 2 · 3 marks

a) Simplify $7m + 2n - 3m$

[1]

b) Simplify $5x \times 2y$

[1]

c) Evaluate $3a - b$ when $a = -2$ and $b = 5$

[1]

3 Expanding & factorising

Lesson 3 · 3 marks

a) Expand $5(y - 2)$

[1]

b) Expand and simplify $3(m + 2) + 2m$

[1]

c) Factorise $15k + 20$

[1]

4 Indices

Lesson 4 · 3 marks

a) Write $3^8 \times 3^2 \div 3^7$ as a single power

[1]

b) Evaluate $2^4 \times 5$

[1]

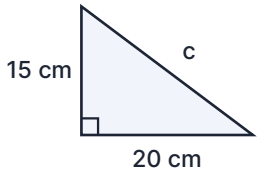
c) Write 252 as a product of prime factors

[1]

5 Pythagoras' theorem

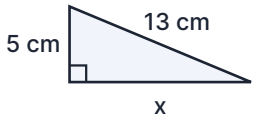
Lesson 5 · 3 marks

a) Find c.



[1]

b) Find x.



[1]

c) Find the hypotenuse if the shorter sides are 5 cm and 9 cm (1 d.p.)

[1]

6 Solve

Lesson 6 · 3 marks

a) Solve $3x + 8 = 29$

[1]

b) Solve $7a - 5 = 3a + 11$

[1]

c) Solve $3x^2 = 75$

[1]

7 Area and volume

Lesson 7 · 2 marks

a) Find the area of a trapezium with parallel sides 6 cm and 10 cm and height 5 cm.

[1]

b) Find the volume of a cylinder with radius 2 cm and height 7 cm (1 d.p.).

[1]

How much did I grow?

QUESTION	LESSON	MY MARKS	ROUNDS I COMPLETED
1 Number skills	Lesson 1	/ 3	Bronze ☐ Silver ☐ Gold ☐
2 Algebra basics	Lesson 2	/ 3	Bronze ☐ Silver ☐ Gold ☐
3 Expanding & factorising	Lesson 3	/ 3	Bronze ☐ Silver ☐ Gold ☐
4 Indices	Lesson 4	/ 3	Bronze ☐ Silver ☐ Gold ☐
5 Pythagoras' theorem	Lesson 5	/ 3	Bronze ☐ Silver ☐ Gold ☐
6 Solve	Lesson 6	/ 3	Bronze ☐ Silver ☐ Gold ☐
7 Area and volume	Lesson 7	/ 2	Bronze ☐ Silver ☐ Gold ☐

Pre-test score: _____ / 20 → Post-test score: _____ / 20

Summary sheet

Number

Same signs $\rightarrow +$, different signs $\rightarrow -$ (for $\times$, $\div$)

$\frac{a}{b} + \frac{c}{d}$: common denominator first

Increase by 15%: $\times 1.15$ · decrease by 20%: $\times 0.8$

Round: next digit 5 or more rounds up

Algebra

Like terms: same pronumerals, same powers

$4a \times 3b = 12ab$, $\dfrac{12xy}{4x} = 3y$

Expand: $a(b + c) = ab + ac$

Factorise: take out the HCF, check by expanding

Index laws

$a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$ $a^0 = 1$

Square root: $\sqrt{49} = 7$ because $7^2 = 49$

Pythagoras

$c^2 = a^2 + b^2$ (c = hypotenuse)

Shorter side: $a^2 = c^2 - b^2$

Triads: $3, 4, 5$; $5, 12, 13$; $8, 15, 17$

Equations

Do the same inverse operation to both sides

Pronumerals on both sides: collect them on one side first

$x^2 = c \Rightarrow x = \pm\sqrt{c}$ · always check

Area and volume

$A = lb$ · $A = \frac{1}{2}bh$ · $A = bh$ · $A = \frac{h}{2}(a + b)$

Kite $A = \frac{1}{2}xy$ · circle $A = \pi r^2$

Prism $V = Ah$ · cylinder $V = \pi r^2 h$

Watch out! Common mistakes

× $3x + 2y = 5xy$ ✓ unlike terms stay apart

× $2(x + 5) = 2x + 5$ ✓ $2x + 10$

× $5^0 = 0$ ✓ $5^0 = 1$

× adding to find a shorter side ✓ subtract

× $-3 - 6 = -3$ ✓ -9

× $2^3 \times 2^4 = 4^7$ ✓ 2^7 (keep the base)

× $x^2 = 36 \Rightarrow x = 6$ only ✓ $x = \pm 6$

× using the slant side as height ✓ perpendicular height

My notes

Coming up in Year 9

Algebra (expanding and simplifying), indices, equations, Pythagoras and trigonometry, financial maths, surface area and volume, data and probability. Every one of them uses the skills in this booklet.